Islamic University of Technology

Board Bazar, Gazipur, Dhaka

SWE 4304 – Software Project Lab-I

Final Project Report

Hall Verse

Hall Management System

March 30, 2026

SPL Team - 14

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Chapter 1
Problem Statement & Potential Solution

1.1 Problem Statement

After moving into the hall, we repeatedly ran into very ordinary but frustrating situations that highlighted the gaps in the existing manual system:

- **Verification headaches** – every time I entered or left the hall the guard would ask for some proof of belonging. If I forgot the document, both sides ended up confused and delayed. Late‑night entries and exits were particularly awkward because there was no quick way for guards to confirm a student's identity.

- **Opaque complaint process** – when something in the hall broke or needed attention, I had to fill out a paper complaint form. The physical process meant complaints often got lost, delayed, or simply forgotten. Students couldn't track the status of their requests, and administrators had no easy way to assign or monitor work.

- **Scattered records** – room allocations, bed statuses, dues payments, and other hall data lived in disparate notebooks or spreadsheets. Updating or retrieving any piece of information required manual searching, which wasted time and introduced errors.

These weren't the result of negligence – they were symptoms of relying on a paper‑based, decentralized system. What we needed was a centralised, digital way to:

- Instantly verify anyone's hall membership,
- Submit and follow up on complaints,
- Manage rooms, beds and payments,
- And facilitate efficient communication between students, guards, and authorities.

1.2 Potential Solution

HallVerse is our answer to those problems. It is a console-based Hall Management System designed to layer seamlessly on top of existing procedures and to make everyday operations smoother, faster and transparent.

The system provides secure **authentication and access control** where students, admins, and workers log in with unique IDs and must change default passwords. A **digital complaint management system** allows students to submit issues, while admins assign them to workers and track statuses until resolved. **Room and bed management** helps admins monitor availability and update occupancy automatically. The system also handles **dues and payments**, enabling administrators to manage fees and penalties efficiently. **Entry and exit logging** records student movements for guards and admins to review. Additionally, **profile and security settings** allow users to update personal information and passwords, while admins can reset student passwords when necessary.

Together, these capabilities create a centralized digital hub – HallVerse – that addresses the verification, complaint tracking, record-keeping and communication problems our motivation described, while preserving the familiar hall environment.

Chapter 2
Implemented Features

The Hall Management System integrates secure authentication, structured information handling, and administrative monitoring to support efficient hall operations. The features are designed to ensure data accuracy, role-based access, and smooth student–authority interaction. The following sections provide an overview of the major functionalities implemented within the system.

2.1 User Authentication System

The user authentication module serves as the primary security system of the application. It verifies user identity and restricts system access based on predefined roles such as Student or Admin. Authentication is performed through a username–password system, with passwords hashed before storage to ensure confidentiality and security. A recovery process is also implemented to assist users who forget their login credentials.

- **Username & Password Authentication**
  - Users log in using valid credentials (Student, Admin or Worker).
  - Passwords are securely hashed before being stored on the CSV files.
  - Password input is masked on the console for added security.
- **Role-Based Access Control**
  - Students, admins, and workers are granted separate levels of access.
  - Students can only interact with student-specific features.
  - Administrators receive privileges to manage system-wide operations.
  - Workers have a dedicated portal to manage assigned complaints.
- **Password Management**
  - Students and workers are required to change their default password after first login.

- Students and workers can update their own password.

2.2 Student Profile Management

This module provides structured management of student information, enabling both students and administrators to handle profile-related data.

- **Student Features**
  - View personal profile information (name, email, room, bed, hall, emergency contact).
  - Update email and emergency contact.
  - View the current hall due.
- **Admin Features**
  - View all student records (ID, name, room, bed, hall, dues).
  - Add new student records with validated inputs.
  - Remove student records; the vacated bed is automatically freed in the room system.
  - Verify student payments by entering an amount to reduce a student's dues.
  - Add penalties for individual students.
  - Apply a hall fee charge to all students at once.

2.3 Complaint & Issue Reporting System

This feature ensures the communication channel between students and hall authorities regarding issues and concerns. Students can submit complaints and track their complaint status, while admins are equipped with tools to process and manage these complaints efficiently.

- **Student Features**
  - File complaints related to electricity, water, room conditions, or other issues.
  - Track complaint status (Pending, In-Progress, Resolved).
  - When a complaint is In-Progress, view the assigned worker's name, service type, work status, and contact number.

- **Admin Features**
    - View all submitted complaints filtered by status (Pending, In-Progress, Resolved), including the student’s name and room number.
    - Assign a worker to any pending complaint; the worker's role is automatically determined from the complaint category.
- **Worker Features**
    - View Complaints assigned to them, filtered by status.
    - Update the status of their assigned complaints.
    - Update their own contact number and password.

2.4 Admin Dashboard

The admin dashboard views all statistics and operational information in a single interface. This feature supports decision-making and enables effective supervision of hall activities.

- **Dashboard Capabilities**
    - Display the total number of registered students, total pending dues, total complaints by status.

2.5 Room Management

A dedicated room management module tracks bed and room availability across the hall.

- **Room Management Capabilities**
    - View all beds filtered by status: All, Occupied, or Vacant.
    - View bed availability by hall or by specific room.
    - Check the status of a specific bed.
    - Reassign a student to a different room/bed.
    - Add new rooms.

2.5 Entry/Exit Management

- Students log their entry or exit through the student menu after login; the system detects the current state (inside/outside) and shows the appropriate action.
- Every entry and exit is recorded with a full timestamp.
- Administrators can view the complete entry/exit log, filterable by student ID.

Chapter 3
Tools & Technologies

In the development of our project, we have selected a set of tools and technologies that ensure smooth collaboration, efficient coding, and reliable version management.

3.1 Version Control

We are using **Git** for version control to track changes in our codebase and manage contributions from all team members. Additionally, **Github** serves as our remote repository which allows us to collaborate, review code, and maintain an organized development workflow.

3.2 Development Environment

For writing and managing our code, we are using **Visual Studio Code**. It provides a lightweight yet powerful environment with extensions that enhance productivity, code navigation, debugging, and C++ development support.

3.3 Programming Language

We will be using **C++** as our primary programming language in this project because **C++** offers strong performance and structured programming capabilities, making it suitable for building a menu-driven console-application like our Hall Management System.

Chapter 4
Application Domain

The **HallVerse** project belongs to the area of Institutional Resource Management, with a specific focus on managing students' information and hall operations. In simple terms, it handles everything required to run a student residential hall smoothly—from assigning rooms to handling maintenance issues.

This system revolves around three main areas. First, there is **residential management**, which includes assigning rooms and beds to students and keeping track of their daily movements in and out of the hall.

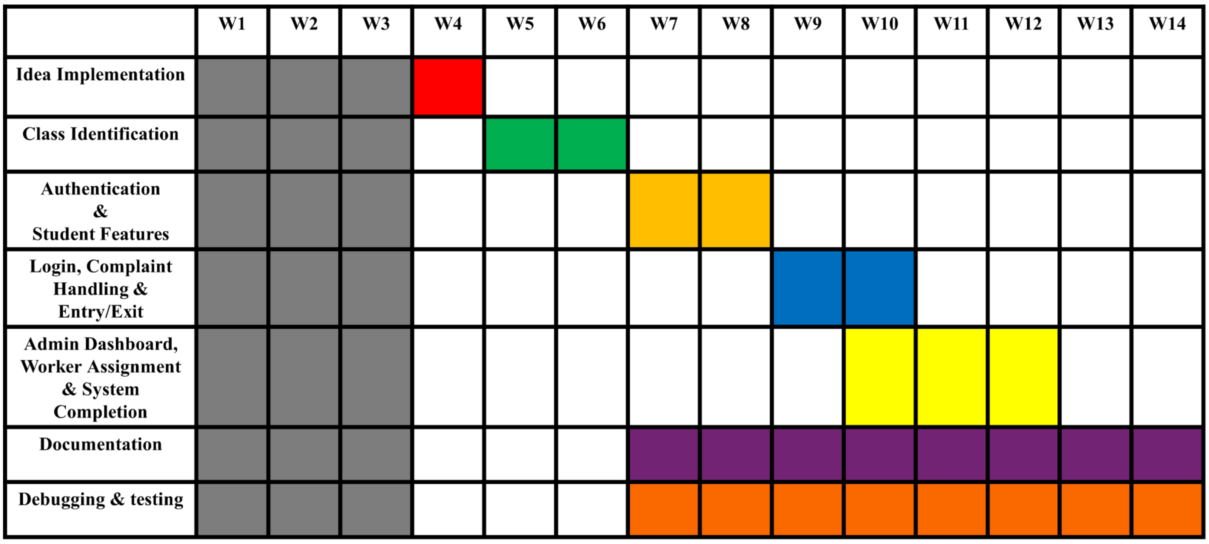
Second, there is **financial management**, where the system ensures that hall fees are collected properly, penalties are applied when necessary, and all dues are tracked accurately from the admin’s end.

Third, the system supports **maintenance and service handling**, allowing students to report issues like electrical or plumbing problems while ensuring that workers are assigned to resolve them efficiently.

The system is designed for three key user groups. **Students** who are the main users who live in the hall need an easy way to report problems. **Administrators** are responsible for managing the system, ensuring data accuracy, and making decisions about resource allocation. **Workers** handle the actual maintenance tasks and rely on the system to receive and complete their assignments.

An important area of this project is its **data-driven nature**. The system must carefully manage relationships between students, their assigned rooms, financial records, and service requests. By digitizing all of this information, the system reduces human error, improves security, and provides a reliable, real-time overview of hall activities—essentially creating a single, trustworthy source of information for everyone involved.

Chapter 5
Project Timeline



Chapter 6
Contribution

Didarul Shahriar (230042137)

- **Project Foundation:** Initialized the project repository, organized the overall structure, and prepared comprehensive documentation
- **User Authentication System (Partial):** Enhanced login handling, implemented password restrictions, email validation, and improved handling of invalid credentials.
- **Student Profile Management**: Designed the Student class and implemented profile viewing, email/emergency contact updates, add/remove student features (Admin-side), and payment verification logic.
- **Complaint & Issue Reporting System (Partial):** Improved the core structure of the Complaint module (complaint.h/.cpp) and contributed to the WorkAssignment manager.
- **Entry/Exit Management (Partial):** Implemented the admin-side interface for viewing complete student entry/exit record logs.
- **Worker Features (Partial):** Initiated the worker workflow and managed the worker.csv data handling.

- **System UI & Utilities:** Developed the InputHelper, implemented screen clearance logic, and refined menu printing and submenu navigation for a smoother user experience.

Muntajim Rahman Saimon (230042103)

- **Core Architecture:** Set up the base header structure and developed the initial UML diagram for the project.
- **User Authentication System (Partial):** Implemented secure password hashing mechanisms and the mandatory password change feature for students and workers upon their first login.
- **Complaint & Issue Reporting System (Partial):** Drove major improvements to the ComplaintManager, enhanced the student complaint viewing interface, implemented worker assignment logic, and designed the tabular view for resolved complaints.
- **Worker Features:** Developed the WorkerManager module and established the core implementation for worker tasks and interaction flows.
- **Admin Dashboard (Partial):** Developed the DashboardManager and implemented the automatic dashboard loading feature upon worker login.
- **Entry/Exit Management (Partial):** Developed the Entry/Exit module and integrated the Date-Time module to ensure accurate timestamp handling for all logs.

Ifham Bin Hossain (230042112)

- **Data & Structure:** Managed the project's file organization and structured CSV-based storage to ensure maintainability and proper data handling.
- **User Authentication System (Partial):** Implemented password validation logic in Main.cpp and developed the password reset mechanism for students.
- **Room Management:** Designed and implemented the complete Room Management module, including bed availability tracking, occupancy filtering, and room-specific status checks.
- **Complaint & Issue Reporting System (Partial):** Initiated the WorkAssignment module and laid the foundation for task allocation logic.
- **Student Profile Management (Partial):** Contributed to the implementation of the Hall Due features.

- **Entry/Exit Management (Partial):** Refined the entry/exit logic by incorporating state-based checks (inside/outside) based on previous log history.
- **Admin Workflow:** Enhanced the admin post-login workflow and improved the system's menu rendering and navigation.

Chapter 7
Future Work

The current project can be transformed from **console-based application** into a **MERN-based web application** to make it more accessible, user-friendly, and scalable.

7.1 Platform Migration to MERN stack

- **Frontend (React):** Implementing a responsive and interactive User Interface (UI) using **React.js,** styled with modern CSS frameworks like **Tailwind CSS.** This will provide a fast, single-page application (SPA).
- **Backend (Node.js & Express):** Developing a robust, non-blocking **RESTful API** to handle full hall management logic and user requests efficiently.

7.2 Database Modernization (MongoDB)

- **Flexible Schema:** MongoDB's document-based structure is ideal for handling diverse student profiles, room allocations, and varying complaint categories.
- **Scalability:** Ensures the system can manage thousands of records and concurrent users without performance degradation.

7.3 Real-Time Notifications & Messaging

- **Instant Alerts (Socket.io):** Integration of **WebSockets**(via Socket.io in the Node environment) to provide real-time updates. Students will receive instant browser notifications when a worker is assigned to their complaint or when their task status changes.
- **Automated Communication:** Integration of **Nodemailer** for automated email alerts regarding pending dues and emergency announcements.

7.4 Enhanced Security & Authentication

- **JWT & Bcrypt:** Replacing simple password checks with **JSON Web Tokens (JWT)** secure, stateless session management, and **Bcrypt** for hashing user passwords.
- **Role-Based Access Control:** Implementing middleware in Express to ensure that Students, Admins, and Workers can only access their authorized features.

7.5 Advanced Functional Features

- **Online Payment Gateway:** Integration of payment gateway like **SSLcommerz** or **Stripe** using the Node.js environments, allowing students to clear hall fees digitally.
- **Digital Entry/Exit with QR Codes:** Implementing a mobile-scannable QR code system at the hall gates to automate "Log Entry/Exit", which will sync directly with the MongoDB database.

Chapter 8
Source Repository

Github Repository: HallVerse